

Week 5 : Analyticity and Integration

Cauchy - Goursat Theorem

We can find a second criteria for the path-independence of contour integrals in terms of the analyticity of the function f .

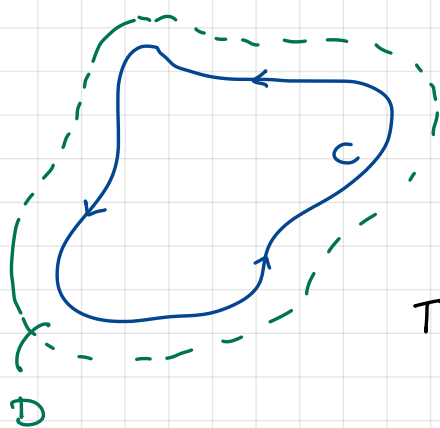
Let C be a simple closed contour oriented positively, and let f be an analytic function on a domain containing the interior of C and the curve C itself.

f is analytic on D and so

$$f(z) = u(x,y) + iv(x,y)$$

and u, v satisfy the Cauchy-Riemann equations.

The contour integral of f can be written in terms of u and v :



$$\begin{aligned}\int_C f(z) dz &= \int_a^b f(z(t)) z'(t) dt \\&= \int_a^b (u(x(t), y(t)) + iv(x(t), y(t))) \cdot (x'(t) + iy'(t)) dt \\&= \int_a^b u(x(t), y(t)) x'(t) - v(x(t), y(t)) y'(t) dt \\&\quad + i \int_a^b u(x(t), y(t)) y'(t) + v(x(t), y(t)) x'(t) dt \\&= \int_a^b u x' - v y' dt + i \int_a^b u y' + v x' dt \\&= \int_C u dx - v dy + i \int_C u dy + v dx\end{aligned}$$

This expression is a differential form, and can be evaluated using Green's Theorem.

In Vector Calculus, Green's Theorem in 2D describes a relationship between the integral over a closed region and its boundary.

Thm: (Green's Theorem 2D)

Suppose L, M are functions of x, y with continuous partial derivatives. Moreover, let C be a closed curve containing a region $\tilde{D}$. Then

anticlockwise
integral $\rightarrow \oint_C L dx + M dy = \iint_{\tilde{D}} \frac{\partial M}{\partial x} - \frac{\partial L}{\partial y} dx dy.$

Applying Green's Theorem to the above:

$$\begin{aligned} \int_C f(z) dz &= \iint_R -\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} dx dy + i \iint_R -\frac{\partial v}{\partial y} + \frac{\partial u}{\partial x} dx dy \\ &= 0 \quad \text{by the Cauchy-Riemann equations.} \end{aligned}$$

However note this argument requires that f be analytic and f' be continuous to apply Green's Theorem.

In fact, using complex analysis we can remove the assumption that f' needs to be continuous. This is important, as we will then be able to prove that the derivative of an analytic function is always continuous (and in fact analytic) using this result.

Thm: Cauchy-Goursat

If f is analytic on a domain D containing a simple closed contour C and its interior then

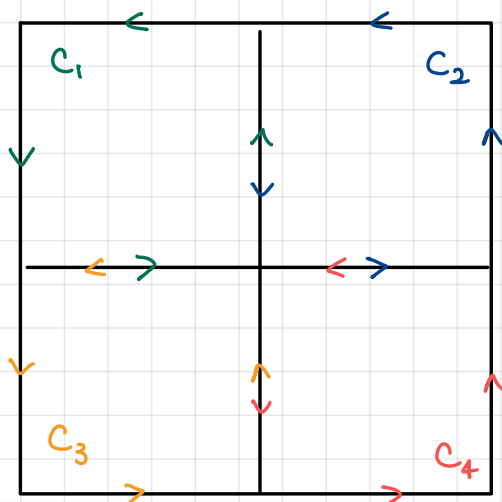
$$\int_C f(z) dz = 0.$$

But why should our complex contour integral depend on the interior region?

This result relies on the two following ideas.

(1) Subdivision of the interior domain.

Example: Suppose we wanted to compute the following sum.



$$I = \sum_{j=1}^4 \int_{C_j} f(z) dz$$

Note that each interior edge is integrated exactly twice but in opposite directions, so their contributions cancel.

What we are left with is only the outer edges.

Hence, if you let $\tilde{C}$ be the contour of the outer edges counterclockwise then

$$I = \int_{\tilde{C}} f(z) dz.$$

(2) Approximation by the derivative

An analytic function has a well behaved derivative, so we want to use this fact to control the behaviour of nearby a point, $z_0 \in \mathbb{R}$.

$$\begin{aligned} f(z) &= f(z_0) + f(z) - f(z_0) \\ &= f(z_0) + f(z) - f(z_0) - f'(z_0)(z - z_0) + f'(z_0)(z - z_0) \\ &= f(z_0) + f'(z_0)(z - z_0) + (z - z_0) \underbrace{\left\{ \frac{f(z) - f(z_0) - f'(z_0)(z - z_0)}{z - z_0} \right\}} \end{aligned}$$

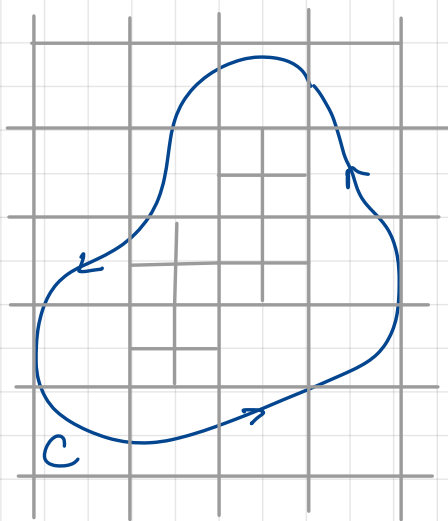
If we can guarantee the final difference is small, then f is very close to linear.

The following Lemma provides a subdivision s.t. the difference above is small on any square.

Lemma: Let f be analytic throughout a closed region R consisting of a positively oriented simple closed contour C and the points it encloses.

For any $\varepsilon > 0$, the region R can be covered by a finite number of squares and partial squares indexed by $j=1, \dots, N$, such that each square has a fixed point z_j for which

$$\left| \frac{f(z) - f(z_j)}{z - z_j} - f'(z_j) \right| < \varepsilon \quad \text{for all } z \text{ in the (partial) square } j.$$



To find the decomposition, we start by subdividing R by 1×1 squares aligned with the real and imaginary axes.

For each square, we try to find a point z_j . If we can, then we move on to the next square.

If we cannot, we subdivide the square into 4 smaller squares by halving the side lengths. Then we repeat.

The theorem states that this procedure will terminate after a finite number of subdivisions.

Proof: By contradiction. Suppose we find a sequence of nested (partial) squares

$$\sigma_1 \supseteq \sigma_2 \supseteq \sigma_3 \supseteq \dots \supseteq \sigma_k \supseteq \dots$$

such that on every square.

$$\forall z_0 \in \sigma_k \text{ and } \forall \varepsilon > 0, \exists w \in \sigma_k \text{ s.t. } \left| \frac{f(w) - f(z_0)}{w - z_0} - f'(z_0) \right| > \varepsilon \quad \text{--- (A)}$$

But we can show that this contradicts the analyticity of f .

Note that each of the squares share at least one common point. Let this point be z_0 . Then since f is analytic at z_0

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ s.t. } \left| \frac{f(w) - f(z_0)}{w - z_0} - f'(z_0) \right| < \varepsilon \quad \forall |w - z_0| < \delta.$$

But then we see for all σ_k contained in the ball $\{w: |w-z_0|<\delta\}$ the property (A) is false. $\downarrow$ □

Proof: Cauchy-Goursat

Let f be analytic in the region R defined by the simple closed contour C and fix $\varepsilon > 0$. Then by the lemma we can decompose R into squares σ_j s.t. $\exists z_j \in \sigma_j$ for which

$$\left| \frac{f(z) - f(z_j)}{z - z_j} - f'(z_j) \right| < \varepsilon \quad \text{for all } z \in \sigma_j.$$

Let C_j be the contour made up of the boundary of σ_j .

Then we know that:

$$(i) \quad \int_C f(z) dz = \sum_{j=1}^N \int_{C_j} f(z) dz$$

$$(ii) \quad \text{If we let } \delta_j(z) = \frac{f(z) - f(z_j)}{z - z_j} - f'(z_j)$$

$$\text{then } f(z) = f(z_j) - z_j f'(z_j) + f'(z_j) z + (z - z_j) \delta_j(z)$$

and by Lemma $|\delta_j(z)| < \varepsilon$ for all $z \in \sigma_j$.

Using (ii) we can find an alternative expression for the integral around C_j :

$$\begin{aligned} \int_{C_j} f(z) dz &= \int_{C_j} f(z_j) - z_j f'(z_j) dz + \int_{C_j} f'(z_j) z dz \\ &\quad + \int_{C_j} (z - z_j) \delta_j(z) dz \\ &= \int_{C_j} (z - z_j) \delta_j(z) dz \end{aligned}$$

because we already know that the functions 1 and z have anti-derivatives.

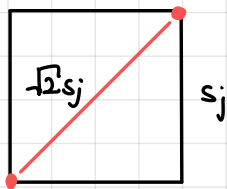
So by (i) .

$$\int_C f(z) dz = \sum_{j=1}^N \int_{C_j} (z - z_j) d_j(z) dz$$

$$\Rightarrow \left| \int_C f(z) dz \right| \leq \sum_{j=1}^N \left| \int_{C_j} (z - z_j) d_j(z) dz \right|$$

To bound the modulus we use the fact that the full region is inside a square of length S and each square σ_j has side length $s_j < S$.

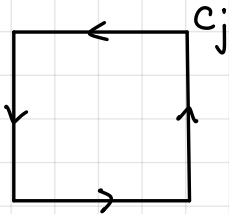
Then



$$\Rightarrow |z - z_j| \leq \sqrt{2} s_j$$

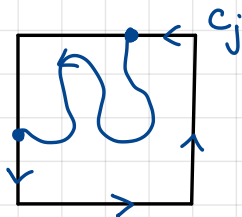
$$\Rightarrow |z - z_j| |d_j(z)| \leq \sqrt{2} s_j \varepsilon$$

and σ_j has a perimeter $4s_j$ and area $A_j = s_j^2$.



Lastly we note if σ_j is a full square then the length of C_j is just the perimeter.

However, if σ_j is a partial square then we can bound the length of C_j by the perimeter of the square plus the length of C found inside the square. Let this length be L_j .



$$|C_j| \leq 4s_j + L_j.$$

So now :

$$\begin{aligned} \left| \int_C f(z) dz \right| &\leq \sum_{j=1}^N \left| \int_{C_j} (z - z_j) d_j(z) dz \right| \\ &\leq \sum_{j=1}^N \sqrt{2} s_j \varepsilon (4s_j + L_j) \\ &= \sum_{j=1}^N (4\sqrt{2} s_j^2 + \sqrt{2} s_j L_j) \varepsilon \\ &\leq \left(4\sqrt{2} \sum_{j=1}^N A_j + \sqrt{2} S \sum_{j=1}^N L_j \right) \varepsilon < (4\sqrt{2} S^2 + \sqrt{2} S L) \varepsilon \end{aligned}$$

Now the constant $4\sqrt{2}S^2 + \sqrt{2}SL$ does not depend on ε or the subdivision. Hence, by making ε small we can make

$\left| \int_C f(z) dz \right|$ as small as we like.

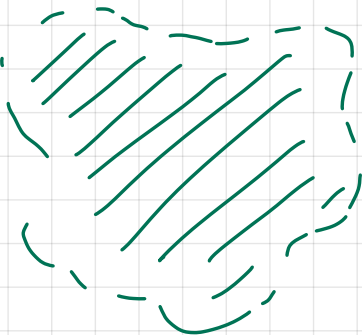
ie. $\int_C f(z) dz = 0.$

□

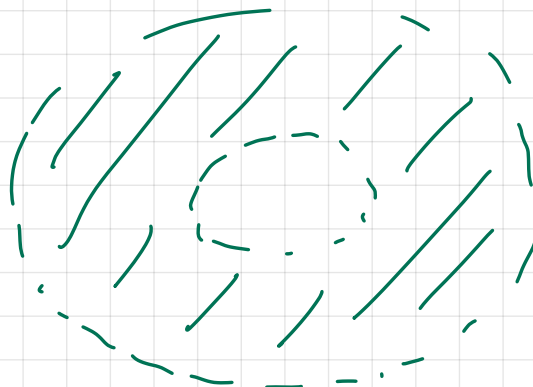
Simply Connected Domains

Def: A simply connected domain D is a domain where any simple closed contour $C \subseteq D$ encloses only points in D .

Eg.



simply connected



not simply connected.

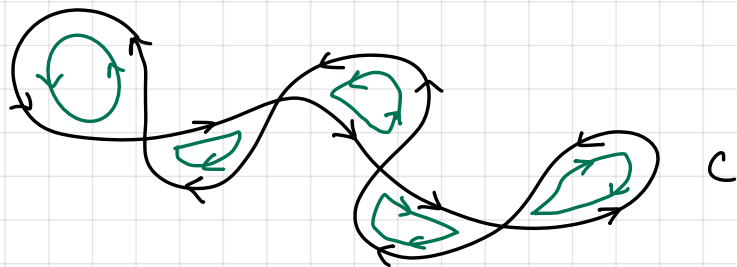
We can generalize the Cauchy-Goursat theorem to non-simple contours when we work on a simply connected domain.

Thm: If a function f is analytic throughout a simply connected domain D , then

$$\int_C f(z) dz = 0$$

for every closed contour C inside D .

Idea: A closed contour C that is not simple can be decomposed into simple closed curves.



So when C decomposes into finitely many pieces, the theorem follows easily from the Cauchy-Goursat theorem.

If an infinite number of curves are needed then the argument is more involved.

Remark: A function f analytic on a simply connected domain D has an anti-derivative on D .

In particular, entire functions always have entire anti-derivatives.

Multiply Connected Domains

A domain which is not simply connected is multiply connected.

Thm: Suppose that

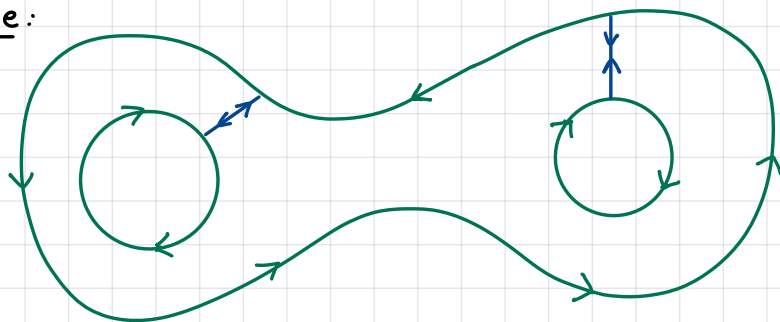
- (a) C is a closed contour, described in the counterclockwise direction, and
- (b) C_k for $k=1, 2, \dots, n$ are simple closed contours interior to C oriented in the clockwise direction, and they are all disjoint and have disjoint interiors.

If a function f is analytic on all the contours C and C_k , and throughout the multiply connected domain consisting on the interior of C and exterior of each C_k , then

$$\int_C f(z) dz + \sum_{k=1}^n \int_{C_k} f(z) dz = 0.$$

Idea: We can add contours to make the multiply connected region into a simply connected one.

Example:



By adding curves joining C to each C_k , we have a single simple closed contour and f is analytic for all interior points. We have also not changed the value of the integrals since we integrate over the new arcs in both directions.

Corollary: Let C_1 and C_2 be positively oriented simple closed contours where C_1 is interior to C_2 .

If f is analytic in the closed region containing C_1 and C_2 and all the points between them, then

$$\int_{C_2} f(z) dz = \int_{C_1} f(z) dz.$$

This property is called the principle of deformation of paths as it shows that if C_1 is continuously deformed into C_2 passing only through the domain of analyticity of f then the value of

$$\int_{C_1} f(z) dz \quad \text{does not change.}$$

This also means, in many cases, we are free to choose the simplest possible contour when evaluating integrals.

It also means that choosing the right contour is very important when looking at problems.

The Cauchy Integral Formula

So far we have seen that analytic functions have anti-derivatives on simply connected domains.

This also means that if analytic functions which are anti-derivatives have analytic derivatives.

In fact, a much stronger result is true.

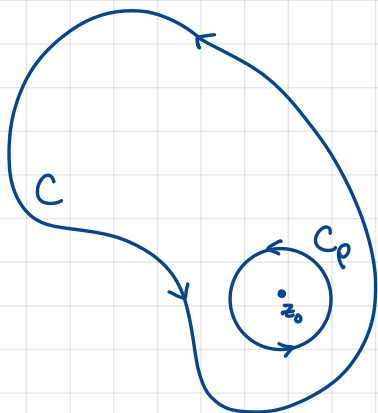
Thm: (Cauchy Integral Formula)

Let f be analytic everywhere inside and on a simple closed contour C oriented counterclockwise. If z_0 is any point interior to C then

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z-z_0} dz$$

That is, when f is analytic on and inside a simple contour then its values inside C are determined completely by its values on C .

Proof: Suppose f is analytic on and inside C and fix z_0 interior to C .



Let C_ρ be the boundary of the ball $\{z: |z-z_0| < \rho\}$ for ρ small enough that C_ρ is inside C .

By deformation of paths:

$$\int_C \frac{f(z)}{z-z_0} dz = \int_{C_\rho} \frac{f(z)}{z-z_0} dz$$

since $g(z) = \frac{f(z)}{z-z_0}$ is analytic for all $z \neq z_0$.

Then

$$\int_C \frac{f(z)}{z-z_0} dz - 2\pi i f(z_0) = \int_{C_\rho} \frac{f(z) - f(z_0)}{z-z_0} dz$$

since
$$\int_{C_\rho} \frac{f(z_0)}{z-z_0} dz = f(z_0) \int_{C_\rho} \frac{1}{z-z_0} dz = 2\pi i f(z_0).$$

We can bound the second integral since by continuity

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } |f(z) - f(z_0)| < \varepsilon \text{ whenever } |z - z_0| < \delta.$$

$$\Rightarrow \left| \int_{C_\rho} \frac{f(z) - f(z_0)}{z - z_0} dz \right| \leq 2\pi\rho \cdot \frac{\varepsilon}{\rho} = 2\pi\varepsilon$$

Since ε is arbitrary, and taking the limit as $\varepsilon \rightarrow 0$ we find:

$$\int_C \frac{f(z)}{z-z_0} dz - 2\pi i f(z_0) = 0$$

□

We can immediately extend this to the following.

Thm: Let f be analytic everywhere inside and on a simple closed contour C oriented counterclockwise. If z_0 is any point interior to C then

$$\frac{d^n}{dz^n} f(z_0) = \frac{n!}{2\pi i} \int_C \frac{f(z)}{(z-z_0)^{n+1}} dz.$$

Corollaries: (1) If a function f is analytic at a given point, then its derivatives of all orders exist and are also analytic.

(2) If $f(z) = u(x,y) + iv(x,y)$ is analytic at a point then u and v have continuous partial derivatives of all orders at that point.

(3) If f is continuous on a domain D and

$$\int_C f(z) dz = 0 \text{ for every closed contour } C \subseteq D$$

then f is analytic throughout D .

(4) Suppose a function f is analytic inside and on a positively oriented circle C_R , centred at z_0 with radius R . If M_R is the maximum value of $|f(z)|$ on C_R then

$$|f^{(n)}(z_0)| \leq \frac{n! M_R}{R^n}.$$